

CHEN XING

PhD Candidate
University of California, Santa Barbara
✉ chenxing@ucsb.edu

🎓 Education

University of California, Santa Barbara

2020 – 2025 (expected)

Ph.D. in Environmental Science & Management

Nanjing University of Information Science and Technology

2013 – 2017

B.S. in Atmospheric Science

🏢 Professional Experience

Graduate Research Assistant

2020 – Present

University of California, Santa Barbara. Advisor: Dr. Samantha Stevenson

Graduate Research Assistant

2017 – 2020

Nanjing University of Information Science and Technology. Advisor: Dr. Fei Liu

📄 Publications

Submitted

- **Xing, C.**, Stevenson, S., Fasullo, J., Harrison, C., Chen, C., Wan, J., Coupe, J., & Pfleger, C. Subtropical Marine Cloud Brightening Suppresses the El Niño–Southern Oscillation. *Submitted to PNAS*.
- **Xing, C.**, Stevenson, S., Di Lorenzo, E., Newman, M., Capotondi, A., Fasullo, J., & Maher, N. Apparent Changes in Pacific Decadal Variability Caused by Anthropogenically-Induced Mean State Modulations. *Submitted to GRL*.

Published

- **Xing, C.**, & Liu, F. (2023). Mount Pinatubo eruption caused the major East China flood in 1991. *The Innovation Geoscience*, 100032–3. [doi:10.59717/j.xinn-geo.2023.100032](https://doi.org/10.59717/j.xinn-geo.2023.100032)
- Liu, F., **Xing, C.**, Chen, L., Gao, C., Lian, T., Zhou, S., et al. (2022). Relative Roles of Land and Ocean Cooling in Triggering an El Niño Following Tropical Volcanic Eruptions. *Geophysical Research Letters*, 49(23), e2022GL100609. [doi:10.1029/2022GL100609](https://doi.org/10.1029/2022GL100609)
- Chai, J., Liu, F., **Xing, C.**, Wang, B., Gao, C., Liu, J., & Chen, D. (2020). A robust equatorial Pacific westerly response to tropical volcanism in multiple models. *Climate Dynamics*, 55(11–12), 3413–3429. [doi:10.1007/s00382-020-05453-6](https://doi.org/10.1007/s00382-020-05453-6)
- Liu, F., **Xing, C.**, Li, J., Wang, B., Chai, J., Gao, C., et al. (2020). Could the Recent Taal Volcano Eruption Trigger an El Niño and Lead to Eurasian Warming? *Advances in Atmospheric Sciences*, 37(7), 663–670. [doi:10/gg9q7h](https://doi.org/10.1007/gg9q7h)
- **Xing, C.**, Liu, F., Wang, B., Chen, D., Liu, J., & Liu, B. (2020). Boreal Winter Surface Air Temperature Responses to Large Tropical Volcanic Eruptions in CMIP5 Models. *Journal of Climate*, 33(6), 2407–2426. [doi:10/gg8wts](https://doi.org/10/gg8wts)
- Zhou, X., Liu, F., Wang, B., Xiang, B., **Xing, C.**, & Wang, H. (2019). Different responses of East Asian summer rainfall to El Niño decays. *Climate Dynamics*, 53(3–4), 1497–1515. [doi:10.1007/s00382-019-04684-6](https://doi.org/10.1007/s00382-019-04684-6)
- Liu, F., **Xing, C.**, Sun, L., Wang, B., Chen, D., & Liu, J. (2018). How Do Tropical, Northern Hemispheric, and Southern Hemispheric Volcanic Eruptions Affect ENSO Under Different Initial Ocean Conditions? *Geophysical Research Letters*, 45(23). [doi:10.1029/2018GL080315](https://doi.org/10.1029/2018GL080315)

Selected Presentations

American Geophysical Union Fall Meeting (upcomming)	December 2024 <i>Washington DC</i>
Apparent Changes in Pacific Decadal Variability Caused by Anthropogenic Forcing-Induced Mean State Modulations <i>Department of Energy's Earth and Environmental Systems Modeling PI Meeting (Oral)</i>	August 2024 <i>Washington DC</i>
Subtropical Marine Cloud Brightening Suppresses the El Niño–Southern Oscillation <i>Gordon Research Conference and Seminar of Climate Engineering (Oral and Poster)</i>	February 2024 <i>Lucca, Italy</i>
Why does Marine Cloud Brightening dampen El Niño–Southern Oscillation <i>American Geophysical Union Fall Meeting (Oral)</i>	December 2023 <i>San Francisco, CA</i>
Why does Marine Cloud Brightening dampen El Niño–Southern Oscillation <i>Solar Climate Intervention Virtual Symposia (Invited Talk)</i>	November 2023 <i>Virtual</i>
Aerosol emissions influence the evolution of North Pacific gyre circulation <i>NOAA Physical Sciences Laboratory Seminar (Invited Talk)</i>	September 2022 <i>Virtual</i>
Aerosols Influence on North Pacific Ocean Variability <i>Ocean Sciences Meeting (Oral)</i>	March 2022 <i>Virtual</i>
Anthropogenic Aerosol Triggered Multi-Decadal North Pacific Ocean Variability Responses in CESM Large Ensemble Project <i>American Geophysical Union Fall Meeting (Oral)</i>	December 2021 <i>New Orleans, LA</i>
Disappearance of Western North Pacific Anomalous Anticyclone in Post-El Niño Summers during Little Ice Age <i>National Postdoctoral Forum on Marine Science and Engineering (Oral)</i>	October 2019 <i>Zhuhai, China</i>
Changes of El Niño-Asian Summer Monsoon Relationship during Last Half Millennium <i>NJU-NUIST Joint Symposium on Climate Dynamics (Oral)</i>	May 2019 <i>Nanjing, China</i>
Boreal-winter Surface Air Temperature Responses to Large Volcanic Eruptions in CMIP5 Models <i>International Workshop on the Decadal-Centennial Variability of Global Monsoon (Oral)</i>	June 2018 <i>Honolulu, Hawaii</i>

Professional Service

Reviewer for Nature Communications	2024
------------------------------------	------

Honors & Awards

H. William Kuni Bren Research Award, UCSB	2022
Bren Fellowship, UCSB	2020
Second-Class Scholarship for Graduate, NUIST	2019
First-Class Scholarship for Graduate, NUIST	2018
Excellent Honor Graduate Award, NUIST	2017
First-Class Scholarship, NUIST	2017
Second-Class Scholarship, NUIST	2015
First-Class Scholarship, NUIST	2014

Technical Skills

Programming Languages: NCL, CDO, Python, FORTRAN, C++, Linux

Techniques: Climate Modeling, Large climate model data, Data Visualization, Statistical Analysis